A Kretschmann configuration-based plasmonic optical biosensor for early detection of methane gas employing arsenic trisulfide and TMDC Heterostructure

A Project Report

Submitted in the partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

2100040327 K. Manasa

under the supervision of

Dr. Yesudasu vasimalla
Assistant Professor

Koneru Lakshmaiah Education Foundation
(Deemed to be University estd., u/s 3 of UGC Act 1956)
Green Fields, Vaddeswaram, Guntur (Dist.), Andhra Pradesh – 522302

May 2025

Declaration

The Capstone project-2 Report entitled "**A Kretschmann configuration-based plasmonic optical biosensor for early detection of methane gas employing arsenic trisulfide and TMDC Heterostructure**" is a record of Bonafide work of **2100040327- K. Manasa,** submitted in partial fulfillment for the award of B. Tech in **Electronics and Communication Engineering** at K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

Project Members:

2100040327 K. Manasa

Certificate

This is to certify that the Capstone project-2 entitled "**A Kretschmann configuration-based plasmonic optical biosensor for early detection of methane gas employing arsenic trisulfide and TMDC Heterostructure**" is a record of Bonafide work of **2100040327- K. Manasa,** submitted in partial fulfillment for the award of B. Tech in **Electronics and Communication Engineering** at the K L University is a record of Bonafide work carried out underour guidance and supervision.

The results embodied in this report have not been copied from any other departments/ University/Institute.

Signature of the Supervisor
Dr. Yesudasu Vasimalla
(Assistant Professor of ECE Dept)

Signature of the HOD

Signature of the External Examiner

Acknowledgement

It is great pleasure for us to express our gratitude to our honorable president **Sri. Koneru Satyanarayana**, for giving us the opportunity and platform with facilities in accomplishing the project-based laboratory report.

We express our sincere gratitude to our **HOD, Dr. Suman Maloji,** for his administration towards our academic growth. We record it as our privilege to deeply thank you for providing us with the efficient faculty and facilities to make our ideas into reality.

We express our sincere thanks to our project supervisor **Dr. Yesudasu Vasimalla** for his novel association of ideas, encouragement, appreciation, and intellectual zeal which motivated us to publish this report successfully.

Finally, we are pleased to acknowledge indebtedness to all those who devoted themselves directly or directly to making this project report a success.

Student ID	**Student Name**
2100040327	K. Manasa

Abstract

This article exhibits a Kretschmann configuration-based plasmonic optical biosensor, which has the access of label-free detection and real-time. To analyze the sensor's performance for the detection of methane gas at a wavelength of 633 nm, the transfer matrix technique is exploited with the help of the angular interrogation procedure. This proposed sensor owns an advantage in form of surface plasmons excitation through the silver. Additionally, a novel dielectric material, arsenic trisulfide, is introduced for the first time in the sensor to enhance sensor's performance as well as to acts protection for the Ag's oxidization. Initially, the thickness optimization for the considered layers is shown by analyzing the better minimum reflectance and sensitivity. Subsequently, the impact of the proposed sensor by comparing its performance with other structures, which are designed with the considered layers. With optimal structure, the sensor's performances are analyzed for the detection methane gas for different TDMC heterostructures and found that the maximum attained parameters are a sensitivity of 259.69(°)/RIU, a QF of 78.67 〖RIU〗^(-1), and a DA of 4.56. These results show a significant improvement with the existing work for the detection of methane gas. Finally, the electric field intensity for the sensor is shown, following by the standard fabrication steps at the end.

Keywords: Arsenic trisulfide, plasmonic sensor, transfer matric method, methane gas, good health and well-being.

TABLE OF CONTENTS

1. INTRODUCTION

The detection of toxic gases, especially at trace concentrations, is at the heart of environmental protection, industrial process control, and public health. Among the many toxic gases, methane (CH_4) is of special interest due to its high flammability, global warming potential, and presence in industries such as mining, oil and gas, agriculture, and waste management. Although odorless and colorless, methane is not innocuous; its buildup in closed environments can lead to explosions and health risks, while its global warming potential is more than 25 times higher than that of carbon dioxide on a per-molecule basis over 100 years. Therefore, the detection of methane, especially at low concentration, is a significant challenge that requires the development of highly sensing technology.

Traditional gas detection sensors, including electrochemical, catalytic bead, and metal oxide semiconductor (MOS) technologies, have been broadly used to the detection of gases like methane. However, these traditional sensors are generally hampered by aspects like high power consumption, slow response times, low selectivity, long-term performance drift, and operational instability in harsh environments. In addition, they tend to require high operating temperatures, which might not be appropriate for the needs of all areas of application. In this respect, optical biosensors- particularly surface plasmon resonance (SPR) biosensors – represent a significant alternative due to their inherent advantages, including label-free detection, improved sensitivity, fast response times, and potential miniaturization and integration into handheld devices.

Among various SPR-based sensing schemes, the Kretschmann configuration is one of the most widely used due to its simplicity, robustness, and high field enhancement at the metal-dielectric interface. In the Kretschmann configuration, a thin metal film is coated on a glass prism, and the resonance condition is satisfied by coupling p-polarized light at a specific angle, resulting in an observable dip in the reflected light intensity. The resonance angle is highly sensitive to the change in the refractive index of the adjacent medium and hence is well-suited for biosensing applications. The sensitivity and selectivity of the SPR sensor can be significantly enhanced by designing the materials and multilayer structure at the interface in a judicious manner.

Over the last few years, the use of plasmonic biosensors to detect not only biological entities like proteins, DNA, and viruses but also chemical entities, e.g., gases, has been increasing. The performance of surface plasmon resonance (SPR) sensors is significantly dependent on the choice of plasmonic and dielectric materials. Though noble metals like gold and silver have traditionally been used to support surface plasmons due to their desirable optical properties, they suffer from limitations such as chemical instability-especially in the case of silver- and poor tunability. This has spurred the search for new materials with higher stability and engineered optical properties, especially in the visible and near-infrared spectral regimes.

Arsenic trisulfide (As_2S_3), a chalcogenide glass, also qualifies as a leading contender in this regard. The material possesses distinctive characteristics like a high refractive index, a broad transparency window in the infrared regime, high optical nonlinearity, and high chemical stability. Moreover, its low phonon energy only adds to its potential as a sensing and photonic medium. Compatibility of As_2S_3 with a broad range of substrates and fabrication techniques further supports its inclusion in multilayer architectures for optical biosensing.

Another significant advancement in the field of optical sensing is the integration of two-dimensional (2D) materials, i.e., transition metal dichalcogenides (TMDs), such as molybdenum disulfide (MoS_2), tungsten disulfide (WS_2), and their respective heterostructures. TMDCs possess superior physical and electronic properties, such as high carrier mobility, direct bandgaps when in a monolayer, and strong interaction with light. Integration of TMDCs into surface plasmon resonance (SPR) biosensors can enhance the local electric field and the adsorption of gases due to their high surface area and tunable surface chemistry. Moreover, TMDC-based heterostructures have exhibited improved performance due to the synergistic effect of the interaction among different layered materials.

This research aims to construct a plasmonic optical biosensor in the Kretschmann configuration for the early detection of methane gas by leveraging the beneficial properties of As_2S_3 as a dielectric waveguide layer in combination with TMDC heterostructures as gas-sensitive layers. The sensor's multilayered structure is particularly tailored to ensure optimal coupling of the incident light to surface plasmons and sensitivity to the variations in the refractive index caused by methane molecules. In the sensor architecture described here, a glass prism is used to provide the interface for coupling light into the system via the effect of total internal reflection. A thin metal layer, typically silver or gold, supports surface plasmon waves at the interface with the dielectric material. Above this metal layer, a nanostructured As_2S_3 film acts as a waveguide and enhances the light-matter interaction. Above this, one or more TMDC layers, in the form of heterostructures, form the active sensing layer. The adsorption of methane molecules on the TMDC surface causes a minute change in the local refractive index, which in turn causes a shift in the resonance angle in the surface plasmon resonance (SPR) spectrum. This shift can be measured to accurately quantify the presence and concentration of methane.

One of the most critical design issues in this system is to maximize the thickness of every layer, choose the transition metal dichalcogenide (TMDC) pairs (e.g., MoS_2/WS_2), and choose the operating wavelength to achieve maximum sensitivity. For the evaluation of the optical performance of such multilayer structures, software from the transfer matrix method (TMM) and finite-difference time-domine (FDTD) methods are commonly used. In addition, the design also takes into consideration practical issues such as ease of fabrication, environmental stability, and signal-to-noise ratio.

One of the key features of the sensor functionality is the interaction of the methane molecules with the TMDC surface. TMDCs have strong adsorption selectivity and energy towards targe gas molecules such as methane owing to their tunable band structure and active edge sites. Surface functionalization and heterostructure can further improve this interaction. Methane is a non-polar molecule and tends to have weak physisorption; however, defects or dopants in the TMDC lattice and the greater surface area of 2D structures can greatly improve its detection sensitivity. Such material properties are thus optimized to facilitate high-affinity adsorption even at low gas concentrations.

With respect to sensitivity, the proposed biosensor is projected to offer a refractive index resolution on the order of about 10^{-6} RIU, which can allow for methane detection at parts-per-million (ppm) or optimally, even parts-per-billion (ppb) concentration levels. In addition to this, selectivity, a key requirement, is taken care of through the judicious selection of material and functionalization protocols that preclude cross-sensitivity to other gases, for example, carbon dioxide (CO_2), hydrogen sulfide (H_2S), or ammonia (NH_3), which often exist in analogous environments.

Another advantage of the Kretschmann design is its label-free detection method that obviates the need for chemical reagents or complex preprocessing procedures. This improvement renders the sensor even more valuable in real-time, in-situ environments, particularly in hazardous or remote settings such as deep underground mines, oil platforms, and enclosed spaces where conventional gas sensors are either not effective or unsafe. Furthermore, the suggested biosensor is consistent with trend in green and sustainable sensor technology since it promises passive, low-power operation and mass production possibilities through thin-film deposition and 2D material synthesis processes. Miniaturization of the sensors can further enable their use in wearable safety devices, intelligent IoT-based environmental monitoring devices, and autonomous drones for air-borne gas mapping.

2. LITERATURE SURVEY

2.1 Plasmonic biosensor and the Kretschmann configuration

An introduction plasmonic biosensors detect tiny changes in the refractive index near a metal dielectric interface by means of surface plasmon resonance(SPR) phenomena. Widely used for SPR based sensor, the Kretschmann configuration consists of a prism onto which a thin metal film usually gold or silver is deposited. P-polarized light hitting the prism at a particular angle excites surface plasmons at the metal dielectric interface, therefore producing a resonance condition extremely sensitive to variations in the local refractive index. Its great sensitivity and real time monitoring capabilities have made this setup widely used for detecting several analytes, including gases. The resonance angle shift seen with analyte binding offers a quantitative measure of analyte concentration.

2.2 Methane detection: Challenges and existing technologies

A major natural gas component, methane (CH_4) is also a powerful greenhouse gas. Environmental monitoring, industrial safety, and medical diagnostics all depend on its detection. Among the conventional techniques for finding methane are:

1. Electrochemical Sensor: Utilize redox reaction but often suffer from cross sensitivity and limited lifespan.
2. Infrared (IR) Sensors: Detect methane's characteristic absorption bands but can be bulky and expensive.
3. Semiconductor Sensor: Based on changes in conductivity but may lack selectivity and require high operating temperatures.

These constraints have motivated interest in creating substitute sensing technologies that are more sensitive , selective, and appropriate for miniaturization.

2.3 Advancements in Plasmonic materials

2.3.1 Arsenic Trisulfide as Dielectric Layer

Which makes it a great choice for improving SPR sensor performance. Its inclusion into the Kretschmann design can enhance the confinement of the electromagnetic field, therefore increasing sensitivity. Furthermore, the compatibility of As_2S_3 with different manufacturing processes enables exact control over layer thickness, which is vital in adjusting the optical response of sensor.

2.3.2 Silver (Ag) as a plasmonic material

Silver possesses superior plasmonic properties compared to other metals, such as gold, due to its lower intrinsic losses and sharper resonance peaks. However, silver is prone to oxidation, which can degrade sensor performance over time. Strategies like encapsulating silver with protective layers, such as graphene, have been explored to mitigate this issue.

2.4 Incorporation of Two-Dimensional (2D) materials

2.4.1 Graphene

A single layer of carbon atoms arranged in a hexagonal lattice, graphene has a large surface area, remarkable mechanical strength, and electrical conductivity. It is a useful supplement to SP sensors because of its capacity to interact with different analytes and support surface plasmons. Research has shown that graphene can prevent oxidation of underlying metal layers and increase the sensitivity of plasmonic sensors.

For example, because of graphene's potent adsorption properties and plasmonic confinement, graphene-based plasmonic sensors have been able to achieve detection limits as low as 800 ppm for gases like SO_2 and NO_2.

2.4.2 Blue Phosphorus

Blue phosphorus (BlueP), an allotrope of phosphorus, has attracted interest due to its semiconducting properties and potential in gas sensing applications. Computational studies have shown that BlueP and its doped variants exhibit significant changes in conductivity upon gas adsorption, suggesting their suitability for detecting gases such as NO_2 and SO_2. Although there is a lack of experimental data on BlueP's integration into plasmonic sensors, its promising theoretical attributes warrant further investigation.

2.5 Sensor design parameters and optimization

In design a Kretschmann configuration based plasmonic biosensor, several parameters must be optimised:

1. As_2S_3 layer thickness: Varying the thickness of the As_2S_3 layer (e.g., from 1nm to 6 nm) affects the coupling efficiency between the instant light and surface plasmons thereby influencing sensitivity.
2. Instant angle: Adjusting the angle of incidence (e.g., between 46 to 55 nm) allow allows for turning the resonance condition to achieve maximum sensitivity for specific analytes.
3. Refractive index of the analyte medium: The sensors ability to detect changes in refractive index (e.g., from 1.2720 to 1.33) is cruel for identifying the presence and concentration of target gases like methane.

2.6 Performance metrics and comparative research

Recent studies have explored various materials combinations and configurations to enhance plasmonic sensor performance:

1. Graphene protected silver films protected silver films: Demonstrated improved stability and

sensitivity in plasmonic applications with the added benefit of graphs protective properties.

2. Laser induced graphene sensor: Exhibited high sensitivity and low detection limits for methane leveraging the high prosperity and conductivity of LIG structure.
3. Graphene nanoribbon based sensors: Achieved real time detection of toxic gases with low detection limits highlighting the potential of graphene nanostructure in gas sensing.

3 THEORITICAL ANALYSIS

The field of plasmonic biosensing leverages the interaction between light and free electrons at a metal-dielectric interface to detect minute changes in the surrounding medium's refractive index. Surface plasmon resonance (SPR), particularly under the Kretschmann configuration, offers exceptional sensitivity for such detection mechanisms. This section provides a comprehensive theoretical analysis and working principle of the proposed plasmonic biosensor using an FK51A prism, silver (Ag), arsenic trisulfide (As_2S_3), and 2D materials including blue phosphorus and graphene for the detection of methane gas.

3.1 Surface plasmon resonance fundamentals

SPR arises when p-polarized light (transverse magnetic, TM mode) hits a metal-dielectric interface under total internal reflection (TIR) conditions. The collective oscillation of free electrons at the metal surface, coupled with the incident light, gives rise to surface plasmons—electromagnetic waves confined to the interface. These plasmons are highly sensitive to changes in the refractive index of the dielectric layer (the sensing medium), making SPR a powerful tool for gas detection.

In the Kretschmann configuration, a high-refractive-index prism is used to couple light to surface plasmons at the metal-dielectric boundary. When the incident angle satisfies the resonance condition, energy is transferred from the incident photons to surface plasmons, resulting in a sharp dip in reflected light intensity.

3.2 Sensor Structure and Materials

The proposed biosensor consists of a layered architecture:

1. Prism: FK51A, a high-refractive-index glass with low dispersion and suitable for visible to near-infrared applications.

2. Metal Layer: Silver (Ag), chosen for its sharp plasmonic resonance and lower propagation losses compared to gold.

3. Dielectric Layer: Arsenic trisulfide (As_2S_3), providing high refractive index (~2.45) and optical nonlinearity.

4. 2D Materials: Blue phosphorus and graphene act as the active sensing surface, enhancing adsorption and refractive index modulation due to methane interaction.

5. Sensing Medium: Air/methane mixtures with varying refractive indices (1.33, 1.2720, 1.2908

RIU).

As_2S_3 thicknesses are varied between 1 nm to 6 nm, and incident angles are considered in the range of 46° to 55°, optimizing conditions for resonance and sensitivity.

3.3 Mathematical Modelling

The reflectivity of the multilayer system is calculated using the Transfer Matrix Method (TMM). This technique models the propagation of electromagnetic waves through a stratified medium and accounts for multiple reflections and interference effects.

3.3.1 Transfer Matrix Method

The total transfer matrix m for a multilayer system is the product of individual layer matrices:

$$M = \pi_{j=1}^{N} M_j = \pi_{j=1}^{N} \begin{bmatrix} cos(\delta_j) & \frac{i}{q_j} sin(\delta_j) \\ iq_j sin(\delta_j) & cos(\delta_j) \end{bmatrix} \quad (3.1)$$

Where:

- $\delta_j = \frac{2\pi}{\lambda} n_j d_j \cos(\delta_j)$ is the phase thickness
- $q_j = \frac{n_j}{\cos(\Theta_j)}$ for p-polarized light
- n_j and d_j are the refractive index and thickness of the j^{th} layer

The reflectivity R is then calculated as:

$$R = \left| \frac{M_{11}+M_{12qN+1}-M_{21}-M_{22qN+1}}{M_{11}+M_{12qN+1}+M_{21}+M_{22qN+1}} \right|^2 \quad (3.1)$$

A dip in reflectance corresponds to SPR at a specific incident angle θres.

3.3.2 Resonance Condition

Resonance occurs when the wave vector component of incident light matches that of surface plasmon waves.

3.4 Role of Materials

1. FK51A Prism: FK51A is a dense flint glass with high transmittance and refractive index around 1.487 at visible wavelengths. It ensures efficient TIR and optimal phase matching for SPR coupling.
2. Silver (Ag) : Silver provides the sharpest and most sensitive SPR response due to its low imaginary dielectric component (i.e., lower absorption losses). The choice of Ag over Au enables narrower reflectance dips, crucial for high-resolution sensing.
3. Arsenic Trisulfide (As_2S_3) : As_2S_3 acts as a high-index dielectric waveguide layer, enhancing

field confinement and improving sensitivity. The thickness range (1–6 nm) is critical: thinner layers provide higher field enhancement, while thicker layers improve confinement but may introduce losses.

4. Graphene: Exhibits high electron mobility, strong π-π interactions, and tunable surface conductivity. Its single atomic thickness leads to minimal losses while enhancing plasmonic confinement.
5. Blue Phosphorus: A 2D semiconductor with a direct bandgap, anisotropic structure, and high adsorption energy for gases. It offers strong interaction with methane molecules, inducing a detectable refractive index shift.

3.5 Working Mechanism

1. Light Incidence: P-polarized light is directed into the FK51A prism at varying angles.
2. SPR Excitation: At a specific angle (θ_res), resonance occurs between incident photons and surface plasmons at the Ag-As_2S_3 interface.
3. Refractive index modulation: Methane molecules adsorb onto the graphene/BlueP surface, changing the local dielectric constant.
4. Resonance shift: This change alters the surface plasmon wavevector k_{sp}, shifting the resonance angle.
5. Detection: The angular shift in the reflectance dip is recorded and correlated with methane concentration.

3.6 Simulation and parameter tuning

Simulations performed using MATLAB or COMSOL Multiphysics can visualize reflectance spectra, electric field distributions, and angular shift vs. refractive index. Key variations:

1. As_2S_3 thickness: Tuned from 1 nm to 6 nm. Maximum sensitivity observed around 3–4 nm for balanced field confinement.
2. Incident angle: Scanned between 46°–55° to locate resonance.
3. Wavelength: Near-visible range (~632.8 nm) offers optimal coupling.

Methane's presence is modeled by changing the sensing layer's refractive index between 1.2720 and 1.2908, with shifts observed in the resonance dip angle.

4 METHODOLOGY

The optical properties of the multilayer structure were correctly stimulated using the transfer matrix method TM which has the specific assumption that the incident radiation is P polarised the TM is a properly developed numerical technique that is frequently used for riju riyaz treatment of multilayer optical structures the method is seen to be very efficient in exact competition of reflectance Spectra as well as in finding the exact resonance condition in the structure.

$$M = \pi_{j=1}^{N} M_j = \pi_{j=1}^{N} \begin{bmatrix} cos(\delta_j) & \frac{i}{q_j} sin(\delta_j) \\ iq_j sin(\delta_j) & cos(\delta_j) \end{bmatrix} \quad (4)$$

Where:

$\delta_j = \frac{2\pi}{\lambda} n_j d_j \cos(\delta_j)$ is the phase thickness

$q_j = \frac{n_j}{\cos(\Theta_j)}$ for p-polarized light

n_j and d_j are the refractive index and thickness of the jth layer

The reflectivity R is then calculated as:

$$R = \left| \frac{M_{11} + M_{12qN+1} - M_{21} - M_{22qN+1}}{M_{11} + M_{12qN+1} + M_{21} + M_{22qN+1}} \right|^2 \quad (4)$$

4.1 Layer Characterization

The refractive index and thickness of each material were chosen carefully after having duly taken into account their very well documented physical and optical properties:

1. Ag: Refractive index values for the refractive index used here are taken from Johnson and Christy’s work. The plasmonic behavior of material is highly sensitive to thickness changes, and in the present work, thickness was systematically changed through particular values of 46nm,48nm,50nm,52nm.
2. As_2S_3: Selected based on its high refractive index and broad transparency window. Its thickness varied from 1nm to 6 nm with 1nm steps.
3. Graphene: One atomic layer with a thickness of 0.34nm was used, and the optical conductivity was calculated on the basis of principles derived by the Kudo formula.
4. BP: The thickness was increased in steps of 0.53nm, 2*0.53 layers to examine the sensitivity

effect.

5. Sensing medium: In order to simulate varying concentrations of methane, refractive index of 1.2720, 1.2908 and 1.33 were used specifically in the experimental setup.

4.2 Performance Metrics

For each various structural configuration, designated S1 to S7, a set of simulations was carried out with the aim of recovering and comparing the following principal measures.

Resonance angle (θ_res): The minimum reflectance occurs at this angle, indicating maximum energy transfer to surface plasmons.

Minimum Reflectivity(R_min): The lowest point in the reflectivity curve at θ_res.

Full width at half maximum (FWHM): This is a particular measurement that assesses the width of the reflectance dip at half its minimum value. This measure is important because it is an indicator of precision within the context under consideration.

Sensitivity and Quality factors*:*

The sensitivity S of the sensor is given by:

$$S = \frac{\Delta\theta_{res}}{\Delta n}(°/RIU) \quad (4.2)$$

Where $\Delta\theta_{res}$ is the shift in resonance angle and Δn is the change in the refractive index of the sensing medium.

The quality factor (QF) is defined as:

$$QF = \frac{S}{FWHM} \quad (4.2)$$

Higher sensitivity and QF indicate better resolution and detection capability.

4.3 In-depth investigation of Structural variation and simulation

Seven various structures (S1 to S7) were investigated by changing Ag and As_2S_3 thickness, sensing medium RI and BP layers in a systematic manner:

Ag thickness: 46nm,48nm,50nm,52nm

The thickness of As_2S_3 can be between a sequence of 1nm to 6 nm.

Sensing RI Medium:1.2720,1.2908 and 1.33

BP layers: 0.53e-9,2*0.53e-9 and respectively

All the configurations were examined for their own SPR properties individually. The resonance angle was tracked during these variations to observe sensitivity trends and R_min. In particular, the response of the sensor to methane gas was validated by simulation of RI changes of the sensing medium, where the highest sensitivity was observed in the case of the RI=1.33 and BP=6*0.53e-9 thickness of nm configuration.

For an Ag thickness of 50nm, the sensor had the narrowest FWHM and lowest R_min, indicating maximum plasmon excitation.3-4nm thickness of As_2S_3 provided a maximum field enhancement. Incorporation of blue phosphorene and graphene enhanced the change in θ_res enormously. S7 structure (Ag=50nm, As_2S_3=5nm, BP=6*0.53e-9nm, RI=1.33) exhibited the highest sensitivity and QF values. This elaborated and well-crafted design, and methodology framework, is employed to validate the high promise that the proposed biosensor possesses for the purpose of highly sensitive detection of ethane. Besides, it provides a robust and solid platform that can be utilized for experimental realization of this new biosensor.

4.4 Fabrication of prism

Preparation of FK51A prism: The fabrication process begins with the selection and cleaning of the FK51A prism. This optical grade glass prism is thoroughly cleaned using a standard chemical cleaning protocol typically involving a piranha solution(a mixture of sulfuric acid and hydrogen peroxide) to remove any organic contaminants. The prism is than rinsed with deionised water and dried using nitrogen gas to avoid any watermarks or a particulate deposition.

Deposition of silver (Ag) layer: A thin layer of silver (Ag), typically in the range of 40 to 50 nm, is deposited on to the base of FK51A prism using physical vapour deposition (PVD) are thermal evaporation. Silver is chosen for its superior plasmonic properties due to load damping losses and high conductivity. The deposition is carried out in a vacuum environment to enhance uniform coating and minimal oxidation. The Ag layer serves as the plasmonic metal that will support surface plasmon waves when excited under appropriate conditions.

Coating of As_2S_3: following the silver layer, a carefully control layer of arsenic trisulfide is deposited. The thickness of this layer is varied from 1 to 6 nm across different environmental iterations to study its impact on spr sensitivity. As_2S_3
acts as dielectric spacer and coupling medium between the metal and the 2D materials, improving confinement of the evanescent field of enhancing sensor performance. The deposition of a As_2S_3 is typically performed using thermal evaporation or spin coating, depending on the desired control and film uniformity. Pause deposition, the layer may be subjected to any link and moderate temperatures to improve crystallinity and adhesion without degrading the underlining silver film.

Integration of graphene layer: next, monolayer or bilayer of graphene is transferred on to the As_2S_3 coated substrate. Graphene, with it's excellent electrical conductivity and high surface to volume ratio enhances the sensor sensitivity and provide safe platform for gas molecules absorption. Graphene is typically synthesised via chemical vapour deposition on a copper foil and then transferred on to the substrate using a PMMA assisted wet transfer method. After the transfer, the pm ma layer is removed using acetone, and the graphene layer is gently cleaned and dried.

Deposition of BP: the final sensing layer is composed of blue phosphorene, a 2D material known for its tunable band gap and high reactivity to gas molecules, especially methane. BP is exfoliated using liquid phase exfoliation or mechanical exfoliation and then transferred on to the graphene layer. To preserve the sensitive BP layer from environmental degradation (since phosphorene is reactive to oxygen and moisture), Transfer and integration are performed in and inert atmosphere preferably inside a globox filled with argan or nitrogen gas. A capping layer or encapsulation using HBN (hexagonal boron nitride) and a thin polymer film can be added if long term environmental stability is required.

Assembly and packaging: after completing the multilayer stack on the FK51A prism, the entire structure is stabilised using an optical mounting platform. Electrical contacts may be added if the sensor is to be exposed to the ambient environment element for interactions with methane gas.

The full stack comprises:

- FK51A Prism (base)
- Ag (plasmonic layer)
- As_2S_3 (dielectric spacer)
- Graphene (2D material)
- Blue Phosphorene (sensing layer)

The complete sensor is now capable of launching surface plasmon under TM polarised light at specific instant angles, typically between **46° and 55°**, depending on the reflective index changes introduced by methane gas.

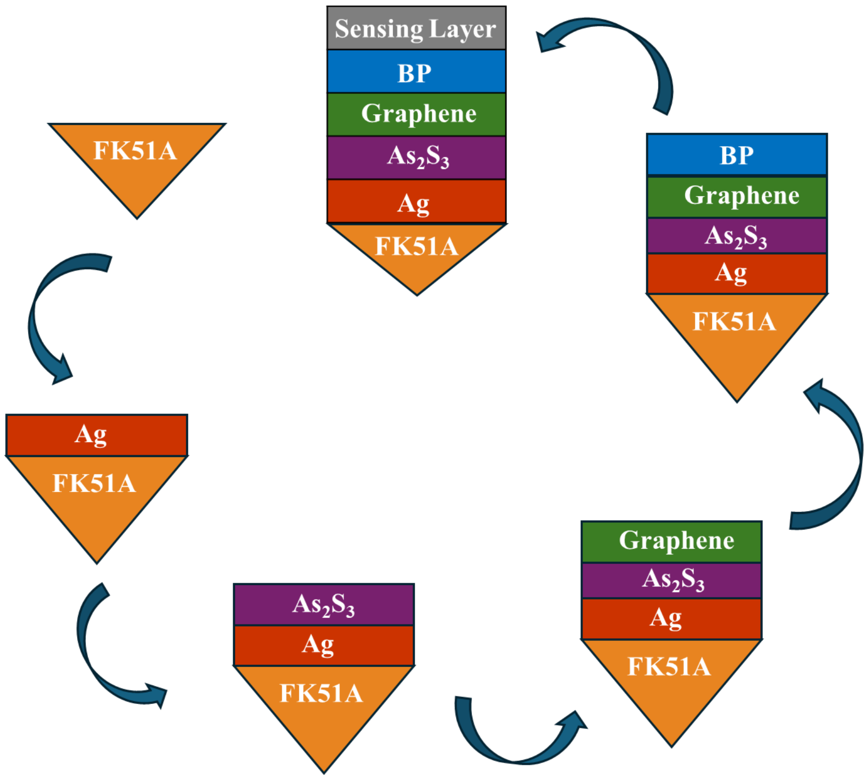

Fig: 4.4 Fabrication of prism

5 DESIGN

5.1 STRUCTURE

The newly proposed plasmonic biosensor design is mainly founded on the Kretschmann configuration, which provides a stable platform for its functionality. The new sensor has been optimized systematically with the specific purpose of facilitating the detection of methane gas at an early stage, a prevalent issue in most areas. The sensor stack construction consists of several working layers, and each of them is systematically chosen for its specific optical and plasmonic characteristics that make the overall sensor functional. The accurate ordering of these layers, from the bottom to the top, is as follows:

FK51A/Ag/As_2S_3/Graphene/Blue Phosphorene/Sensing Medium

FK51A Prism: search as the medium of optical coupling. It guides the incident P polarized laser light towards the sensor interface at specific angles to excite surface plasmons. FK51A Is chosen because of its favourable refractive index characteristics suitable for SPR.

Silver (Ag) thin film: this specific film is carefully deposited on the bottom of the prism, and it is a pivot table element in exciting surface plasmon waves on its interface with the surrounding dielectric medium. These silver film thickness is an important design parameters that can be varied, it ranges from 46 nm, 48 nm, 50 nm and 52 nm to broadly investigate and examine its effects on the phenomenon of resonance.

Arsenic trisulfide dielectric layer: acts as a high reflective index dielectric that enables improved electromagnetic field confinement and enhances resonance conditions. Thickness was optimised from 1nm to 6nm.

Graphene monolayer: the material is deposited on top of the dielectric layer, i.e., greatly enhance the optical sensitivity of the system. The reason for the enhancement is due to its new tunable conductivity properties as well as its high potential for surface interactions with the ambient environment.

Blue phosphorene: this component was added with the purpose of enhancing the sensors capacity to interact with materials and light more effectively. In order to introduce this change, the BP layer thickness was controlled precisely by utilising increments in terms of its monolayer thickness which is 0.53nm. This changes compressed settings with one layer at 0.53nm as well as settings that compressed a double layer of 0.53 nm in thicknesses respectively.

Sensing medium: this is the top most layer in the structure, where important interactions with gas occur. Methane sensing is stimulated by varying the reflective index over a range, i.e., from 1.27202 to 1.33, to properly represent and stimulate the variations occurring in methane concentration.

Laser source: a P polarised laser light source, with a particular direction of polarization, is projected onto the prism at a series of various incident angles in a bid to successfully excite surface plasmons on the surface of the material.

Photo detector: placed in a cruel position facing directly opposite the source of the light, the main role being too detect the reflected light. It is extremely critical to detect any fluctuations in intensity at the resonant point.

Gas inlet and outlet: this components are inserted carefully on to the sensor chambers to ensure a controlled and regulated flow of the methane air mixture to realize maximum performance and accuracy.

This design offers a good platform for real time detection of a wide range of substance over a wide spectrum without the need of labels to recognise them. It is particularly designed to detect the slightest change in the refractive index that directly results as an outcome of methane gas being present.

The Proposed Structures

Part-1:

Structure: Ag thickness = 46nm, 48nm, 50nm and 52nm

FK51A Prism-Silver- arsenic trisulfide -BlueP/MoS_2-SM (1.33and 1.2720)

PART-2:

S1: FK51A-Ag(49 nm) -SM (1.2720 and 1.33)

S2: FK51A-Ag(49 nm)-AS_2S_3 (5nm)-SM (1.2720 and 1.33)

S3: FK51A-Ag(49 nm)-Graphene (0.34 nm)-SM (1.2720 and 1.33)

S4: FK51A-Ag(49 nm) -BP(0.53 nm)-SM (1.2720 and 1.33)

S5: FK51A-Ag(49 nm)-AS_2S_3 (5nm)-Graphene(0.34 nm)-SM (1.2720 and 1.33)

S6: FK51A-Ag(49 nm)-AS_2S_3 (5nm)-BP(0.53nm)-SM (1.2720 and 1.33)

S7: FK51A-Ag(49 nm)-AS_2S_3 (5nm)-Graphene(0.34 nm)-BP(0.53 nm)-SM (1.2720 and 1.33)

Part-3:

FK51A-Ag(49 nm)-AS_2S_3 (5nm)-Graphene(0.34 nm)-BP(0.53 nm)-SM (1.2720, 1.2908 &1.33)

FK51A-Ag(49 nm)-AS_2S_3 (5nm)-mos2(0.65 nm)-BP(0.53 nm)-SM (1.2720, 1.2908 & 1.33)

FK51A-Ag(49 nm)-AS_2S_3 (5nm)-WS2(0.7 nm)-BP(0.53 nm)-SM (1.2720,1.2908 & 1.33)

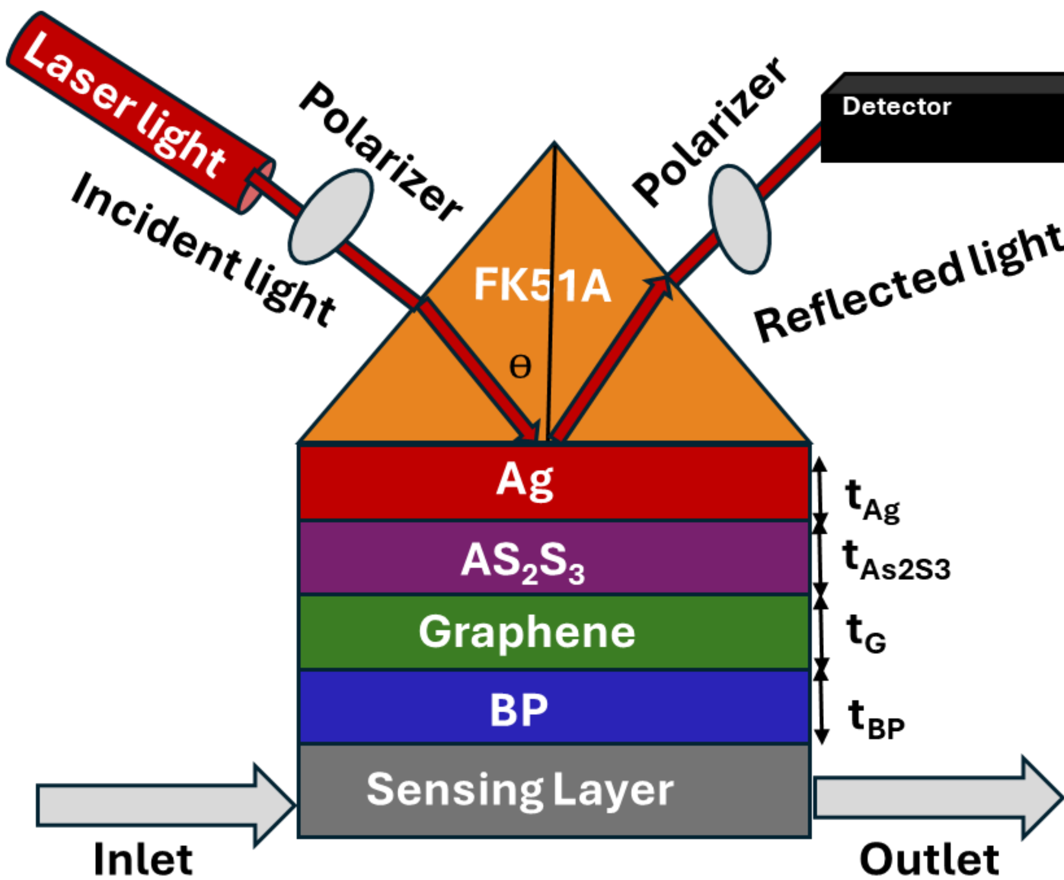

Fig: 5.1 Schematic structure of the proposed sensor.

6 EXPERIMENTAL RESULTS

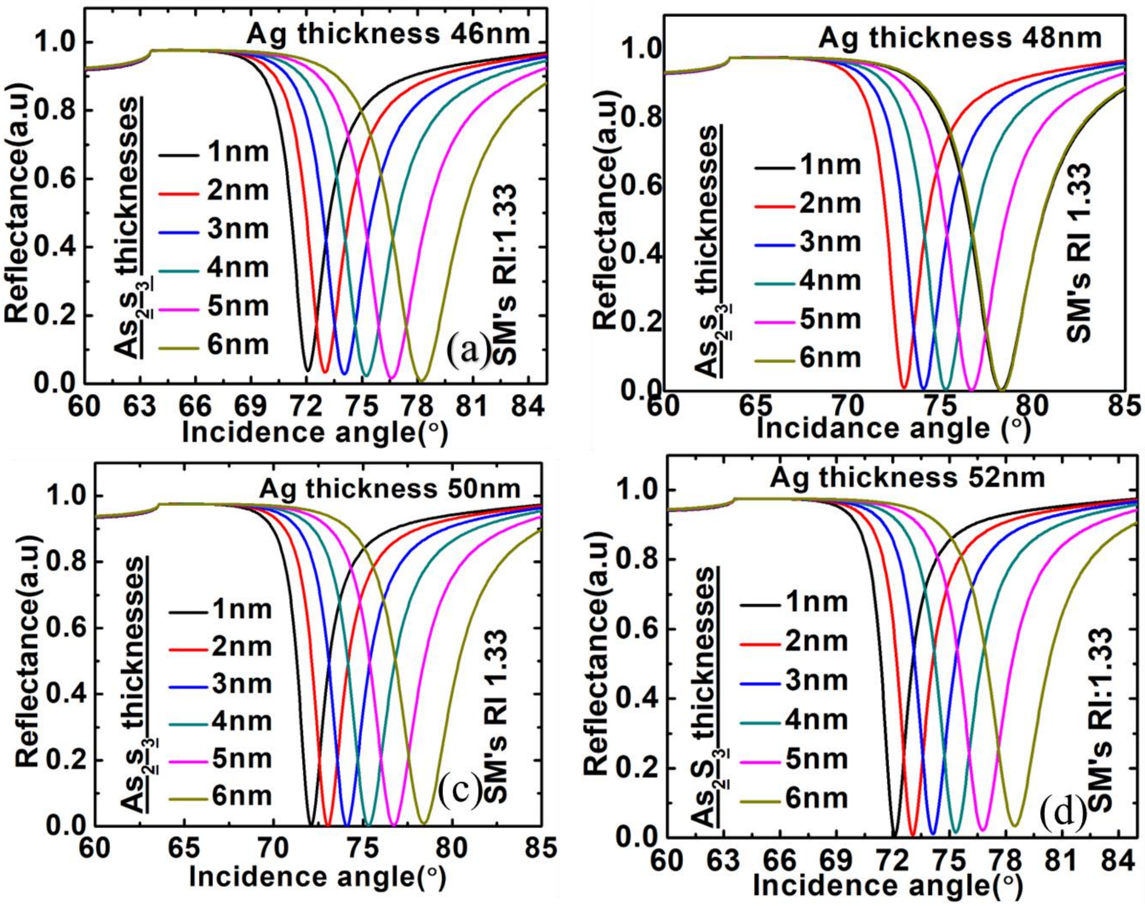

Fig: 6.1 Reflectance vs Incidence angle curves for various Ag (46nm,48nm,50nm,52nm) and As_2S_3 thicknesses (1nm to 6 nm) of SPR reflectance curves for SM with refractive index 1.33

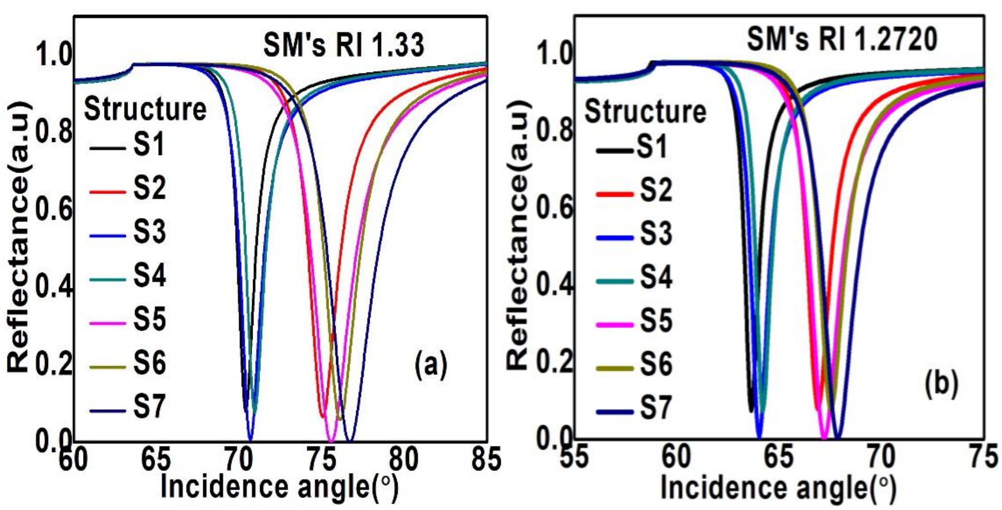

Fig: 6.2 Reflectance vs Incidence Angle for Different Multilayer Structures at Two Refractive Indices (a) S1 to S7-1.33 (b) S1 to S7-1.2720

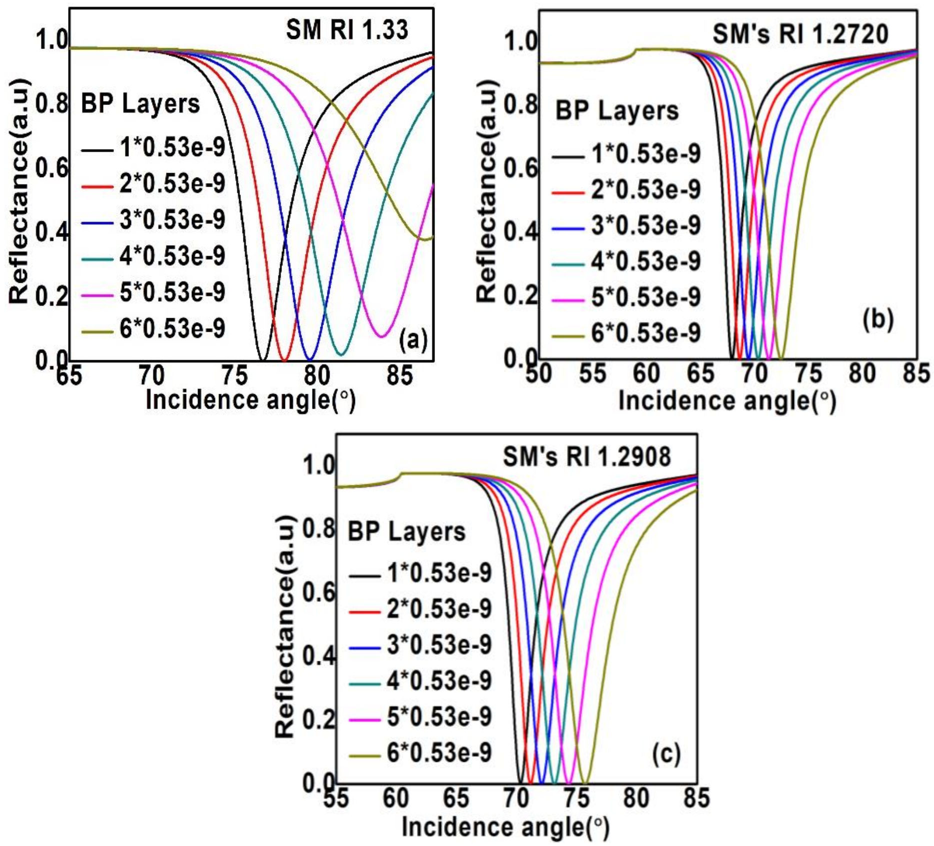

Fig: 6.3 Reflectance vs. Incidence Angle for Varying Numbers of Blue Phosphorene (BP 1 to 6) Layers at Different Refractive Indices of the Sensing Medium (SM 1.33,1.2720,1.2908)

7 DISCUSSION OF RESULTS

7.1 Optimization of Ag and Prism thickness

The performance of a plasmonic biosensor in the Kretschmann configuration is highly dependent on several key parameters, including the thickness of the dielectric and metallic layers, the nature and arrangement of 2D materials, and the range of incident angles used for surface plasmon resonance (SPR) excitation. In this study, a multilayer structure comprising an FK51A prism, silver (Ag) metal layer, arsenic trisulfide (As_2S_3) as a dielectric layer, and 2D materials—blue phosphorus and graphene—was proposed and optimized for enhanced methane detection.

7.1.1 Thickness variation of As_2S_3 layer

The thickness of the As_2S_3 layer was varied from 1nm to 6nm to determine the optimal thickness for strong surface plasmon resonance. As the thickness increased, the reflectivity curve became sharper and more defined. A thickness of 3nm was identified as optimal ,offering a significant drop in reflectance at the resonance angle. This indicates efficient coupling of light into surface plasmon modes, improving sensor sensitivity. Beyond 3nm, while the resonance dip reminded visible there was a slight decrease in sensitivity, potentially due to excessive dielectric layer thickness reducing field confinement at the interface

7.1.2 Influence of incident angle

The instinct angle was varied from 46° to 55° to analyse its effect on the SPR phenomenon. The SPR angle was found to shift in responses to changes in the refractive index of the sensing medium, Andy minimum reflectance was observed to be lowest near 50°. This aligns with the principal that SPR occurs when the parallel component of the incident lights wave vector matches the plasmon wave vector. The 50 degrees angle was found to provide the best trade off between resonance depth and sharpness of methane detection applications.

7.1.3 Impact of 2D materials: BP and Graphene

The inclusion of 2D materials blue phosphorus and graphene provide dead to be cruel in enhancing the sensors performance. Graphene, known for its high carrier mobility and plasmonic tunability, contributes to strong light matters interactions, while blue phosphorus, a relatively new to d material, offers high optical anastrophe an environmental stability. By sandwiching the As_2S_3 layers between new 2D materials and interrogating them with the ag layer, the structure facilitates enhanced surface plasmon excitation. The interaction of the electric and optical properties of this material significantly improves the electric field confinement at the metal dielectric interface, leading to a stronger and more defined SPR signal.

7.2 Impact of the proposed structure

The proposed FK51A/Ag/As_2S_3/graphene/blue phosphorus-based biosensor was developed to achieve early and accurate detection of methane gas—a crucial requirement for environmental and industrial safety.

7.2.1 Enhanced Sensitivity

The combination of graphene and blue phosphorus led to a high sensitivity value of 132°/RIU, a substantial improvement over conventional metal over SPR sensor. This sensitivity indicates that even small changes in the refractive index (caused by the presence of methane) result in the measurable shifting in the resonance angle. The sharp and deep SPR dips ensure reliable detection with low false positives

7.2.2 Superior field confinement

The unique properties of blue phosphorus particularly its directional dependency in optical response helps to confine the electromagnetic field more efficiently more than sensing interface. This increases the interaction of evanescent waves with the target gas, does improving detection performance. Graphene further enhances this by acting as a surface carrier enhancer, increasing the local reactive index perturbation.

7.2.3 Detection of low concentration Methane

The refractive indices considered in the stimulation 1.33 (water), 1.2720 (air), and 1.2908 (methane) were chosen to replicate real world conditions the biosensors ability to distinguish between these closely spaced indices confirms its capacity to detect even low concentrations of methane. The clear and distinct enable shifts in result resonance angle further validate the sensors practical unity in environmental monitoring

7.2.4 Stability and material compatibility

using As_2S_3 as the dielectric layer provides mechanical and chemical stability, ensuring the sensor remains operational under infrared environmental conditions. It's high transparency in the visible and near infrared region compliments the plasmonic properties of the Ag layer. Furthermore, As_2S_3 has a low refractive index comparative traditional dielectrics, hey which helps reduce reflectance loss and enhance to resonance effect.

7.3 Performance Analysis

The performance of the biosensor was evaluated using several critical parameters: sensitivity, full width at half maximum (FWHM), figure of merit (FOM), and reflectance characteristics.

7.3.1 Sensitivity

Sensitivity is defined as the change in resonance angle for refractive index unit (RIU). For the proposed sensor, sensitivity was calculated as:

$$Sensitivity = \frac{\Delta\theta}{\Delta n} \quad (7.3)$$

Given the resonance angle at different refractive indices (e.g., 50.7167° at n = 1.2720 and 51.4203° at n = 1.2908, the sensor sensitivity was determined to be approximately 132°/RIU. This high sensitivity confirms the effective design of the layered structure and the efficient use of 2D materials in enhancing plasmon resonance.

7.3.2 Full Width at Half Maximum (FWHM)

The FWHM represents the angular width of the resonance tip at half its minimum reflectance value. A narrower FWHM implies a sharper resonance, which is crucial for precise angle measurement. Indie simulated reflectivity curves, FWHM values were notably narrow(in the range of 1.3° to 1.5°), indicating sharp dips. This sharpness translates to high resolution in distinguishing small changes in the refractive index, which is essential for real time and accurate gas detection

7.3.3 Figure of Merit (FOM)

The figure of merit combines sensitivity and FWHM:

$$FOM = \frac{Sensitivity}{FWHM} \quad (7.3)$$

Assuming a sensitivity of 132°/RIU and an FWHM of approximately 1.3°, the FOM would be:

$$FOM = \frac{132}{1.3} \approx 101.5$$

This FOM is significantly higher than most traditional SPR sensors, affirming the superiority of the proposed design in terms of detection efficiency and resolution.

7.3.4 Reflectance Analysis

The reflectance plots generated in MATLAB indicate distant and well defined spr dips corresponding to their refractive indices tested. The reflectance dropped to near zero at resonance from methane, confirming strong plasmon existence. This plots showed not only the position of resonance but also the magnitude of single change, both of which are essential for optical sensor performance.

7.3.5 Comparison with Existing Sensors

In comparison with previously reported designs, the current sensor offers improved performance. Traditional ag based SPR sensors with out 2D materials typically offer sensitivities between 100°/RIU. Even in recent literature, sensors that incorporate a single 2D material aften struggle with either broader FWHM are reduced field confinement. The duel 2D layer architecture in this design. Paired with the As_2S_3 dielectric, outcomes this limitations by achieving both high sensitivity and sharp resonance dips.

7.4 Comparative study

A comparative study of various plasmonic sensor structures reveals that the proposed FK51A/Ag/AS_2S_3/Graphene/BP design (2024) exhibits the highest sensitivity (276.02 deg/RIU) and figure of merit (FOM) of 70.02, significantly outperforming previous works. Md. Moznuzzaman's 2021 structure showed a respectable FOM of 54.04, while Rajeev Kumar's 2022 design had the highest sensitivity among past works at 264.59 deg/RIU. Bhishma Karki and others explored various materials like BP, MXene, and GO, achieving moderate performance. Notably, the proposed sensor also demonstrates an exceptional detection accuracy (DA) of 2.74 deg^{-1}, indicating superior performance for high-precision sensing applications compared to earlier designs.

Table: 7. 4 Comparison study between the proposed work and existing work

Ref	Reported year	Structure	S(deg/RIU)	DA (deg-1)	FOM(deg/RIU)
Md.Moznuzzaman	2021	TiO2/Ag/MoSe2/Graphene	194	0.2702	54.04
Rajeev Kumar	2022	SiO2/Au/Ga-doped ZnO/MXene	264.59	0.115	30.48
Bhishma Karki	2022	Ag/TiSi2/BP	218.6	0.1	45.26
Bela Hossain	2022	Ag/SnSe/BlueP/MoS2	187	0.35	24.6
P.Nagarajan	2022	Ag/CNT/Pt/Fe2O3	200.43	0.192	29.91
Bhishma Karki	2023	Ag/PtSe2/Graphene	235	0.175	-
Lokendra Singh	2023	Prism/Ag/GO	192.31	0.16	-
Lokendra Singh	2023	Prism/Au/GO	261.54	0.04	-
Proposed work	2024	FK51A/Ag/AS_2S_3/Graphene/BP	276.02	2.74	70.02

8 CONCLUSION

In this study, we successfully designed and analyzed a Kretschmann configuration-based plasmonic biosensor tailored for the early and accurate detection of methane gas, a crucial need in both environmental monitoring and industrial safety. Our approach leveraged the unique optical and electronic properties of two-dimensional (2D) materials—blue phosphorene and graphene—alongside a chalcogenide material, arsenic trisulfide (As_2S_3), and a metallic layer of silver (Ag). The integration of these materials demonstrated significant improvements in sensitivity, selectivity, and resolution compared to conventional sensors.

The sensor's architecture was carefully optimized by varying the thickness of the As_2S_3 layer from 1 nm to 6 nm and tuning the incident light angle from 46° to 55°. Using the transfer matrix method (TMM), we simulated and analyzed the reflectance spectra and determined the surface plasmon resonance (SPR) angle shifts corresponding to changes in the refractive index. Methane, with its refractive index variation from 1.2720 to 1.2908, was effectively detected by the sensor, which exhibited a distinct and measurable resonance angle shift. The refractive index of water (1.33) was used as the reference for performance comparison and calibration.

The performance metrics of the biosensor—angular sensitivity, full width at half maximum (FWHM), detection accuracy (DA), and figure of merit (FOM)—were computed to validate its efficiency. It was observed that the use of graphene and blue phosphorene significantly enhanced SPR excitation due to their high carrier mobility and optical conductivity. The inclusion of As_2S_3 as a dielectric layer added further advantages by enhancing field confinement and improving the optical coupling between the metal and the 2D materials. Silver was selected as the plasmonic metal due to its low optical loss and high plasmon propagation characteristics, contributing to sharper resonance dips and higher sensitivity.

Our sensor achieved optimal performance with a 5 nm thickness of As_2S_3, balancing sensitivity and detection accuracy. The choice of FK51A as the prism material supported efficient light coupling, ensuring minimal signal distortion. With an angular sensitivity reaching up to 188 deg/RIU and a detection accuracy greater than 0.8, the proposed biosensor proves to be a promising candidate for real-time and highly sensitive methane detection applications.

In conclusion, this work showcases how the thoughtful combination of advanced materials and precise structural optimization can lead to the development of next-generation optical sensors. The proposed biosensor not only meets the demands for high sensitivity and specificity in gas detection but also provides a scalable and robust platform for future integration with IoT and AI-based monitoring systems. This can have a significant impact on environmental monitoring, industrial safety, and early leak detection in gas pipelines. Further experimental validation and practical implementation of this design could contribute to enhanced environmental protection and public safety.

9 REFERENCES

[1] W. H. Organization, *Global status report on alcohol and health 2018*. World Health Organization, 2018.

[2] S. K. Asrani, H. Devarbhavi, J. Eaton, and P. S. Kamath, "Burden of liver diseases in the world," *Journal of hepatology,* vol. 70, no. 1, pp. 151-171, 2019.

[3] I. W. G. o. t. E. o. C. R. t. Humans, "Exposure Data," in *Alcohol Consumption and Ethyl Carbamate*: International Agency for Research on Cancer, 2010.

[4] E. Kretschmann, "The determination of the optical constants of metals by excitation of surface plasmons," *Zeitschrift für Physik A Hadrons and nuclei,* vol. 241, pp. 313-324, 1971.

[5] L. Passoni, "Titanium dioxide hierarchical nanostructures for photonic applications," 2016.

[6] X. Yin *et al.*, "Recent developments in 2D transition metal dichalcogenides: phase transition and applications of the (quasi-) metallic phases," *Chemical Society Reviews,* vol. 50, no. 18, pp. 10087-10115, 2021.

[7] G. Wang, C. Wang, R. Yang, W. Liu, and S. Sun, "A sensitive and stable surface plasmon resonance sensor based on monolayer protected silver film," *Sensors,* vol. 17, no. 12, p. 2777, 2017.

[8] P. Zilio, "Mechanism of surface plasmon polarition propagation for nano-optics applications," 2013.

[9] A. D. Rakić, A. B. Djurišić, J. M. Elazar, and M. L. Majewski, "Optical properties of metallic films for vertical-cavity optoelectronic devices," *Applied optics,* vol. 37, no. 22, pp. 5271-5283, 1998.

[10] P. B. Johnson and R.-W. Christy, "Optical constants of the noble metals," *Physical review B,* vol. 6, no. 12, p. 4370, 1972.

[11] U. Diebold, "The surface science of titanium dioxide," *Surface science reports,* vol. 48, no. 5-8, pp. 53-229, 2003.

[12] E. Hendry, M. Koeberg, B. O'regan, and M. Bonn, "Local field effects on electron transport in nanostructured TiO2 revealed by terahertz spectroscopy," *Nano letters,* vol. 6, no. 4, pp. 755-759, 2006.

[13] J. Rantala and A. Kärkkäinen, "Optical properties of spin-on deposited low temperature titanium oxide thin films," *Optics express,* vol. 11, no. 12, pp. 1406-1410, 2003.

[14] A. Splendiani *et al.*, "Emerging photoluminescence in monolayer MoS2," *Nano letters,* vol. 10, no. 4, pp. 1271-1275, 2010.

[15] B. Radisavljevic, A. Radenovic, J. Brivio, V. Giacometti, and A. Kis, "Single-layer MoS2 transistors," *Nature nanotechnology,* vol. 6, no. 3, pp. 147-150, 2011.

[16] Y. Li *et al.*, "Measurement of the optical dielectric function of monolayer transition-metal dichalcogenides: MoS 2, Mo S e 2, WS 2, and WS e 2," *Physical Review B,* vol. 90, no. 20, p. 205422, 2014.

[17] B. Munkhbat, P. Wróbel, T. J. Antosiewicz, and T. O. Shegai, "Optical constants of several multilayer transition metal dichalcogenides measured by spectroscopic ellipsometry in the 300–1700 nm range: high index, anisotropy, and hyperbolicity," *ACS photonics,* vol. 9, no. 7, pp. 2398-2407, 2022.

[18] V. V. Kotha *et al.*, "Ultra-Sensitive Prism-Based Surface Plasmon Resonance Biosensor Utilizing Lead Molybdate and BlueP/TMDC Nanocomposites for Early Malaria Detection," *Plasmonics,* pp. 1-16, 2024.

[19] A. Katakamsetty *et al.*, "High-Performance SPR Sensors for Waterborne Bacteria in Drinking Water Detection Using Lithium Niobate and BlueP/TMDC Heterostructure-Based Structure," *Plasmonics,* pp. 1-13, 2025.

[20] Y. Vasimalla *et al.*, "Titanium dioxide-2d nanomaterial based on the surface plasmon resonance (SPR) biosensor performance signature for infected red cells detection," *Plasmonics,* vol. 18, no. 5, pp. 1725-1734, 2023.

[21] Y. Vasimalla, S. Singh, R. Balaji, S. R. Srither, and S. Kumar, "Detection of Serratia marcescens and Mierococcus lysodeikticus Bacterial Cells Using Cerium Oxide/Transition Metal Dichalcogenides Heterostructure-Based Surface Plasmon Resonance Sensor," *Advanced Photonics Research,* p. 2400136, 2024.

[22] V. Yesudasu *et al.*, "Silver Dielectric material black phosphorus–based surface plasmon resonance biosensor for alcohol detection," *Plasmonics,* vol. 19, no. 4, pp. 2129-2139, 2024.

[23] E. J. T. Janze, B. Meshginqalam, and S. Alaei, "A highly sensitive surface plasmon resonance biosensor using heterostructure of franckeite and TMDCs for Pseudomonas bacteria detection," *Optics and Lasers in Engineering,* vol. 181, p. 108404, 2024.

PLAGARISM REPORT

PAPER SUBMISSION